

Lab 3 — Peer Review Record

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Peer reviewer: Punnapob Wirojwongchai — 67070503425 — SaintCrois

Pull Requests I authored (reviewed by my partner)

PR	Branch	Reviewer verdict
#44	document/lab3	Approve
#46	feature/lab3-data-model	Approve
#48	feature/12-lab3-auth	Approve
#49	feature/14-lab3-staff-ticket-management	Approve
#50	feature/lab3-admin-users	Approve
#51	feature/16-lab3-integration-qa	Pending

[#44](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/44

[#46](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/46

[#48](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/48

[#49](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/49

[#50](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/50

[#51](#) https://github.com/bravefe/cpe334_lab01_67070503427/pull/51

#44 document/lab3

Issue 10

Branch: document/lab3

Scope

* Author and approve:

* docs/lab-03/specification.md

* docs/lab-03/ui-spec.md

* docs/lab-03/api-spec.md

* docs/lab-03/tests.md

* Create Issues #36 - #42 on the board.

* Confirm the engineering contract is completed before implementation PRs are opened.

Definition of Done

* All four Lab 3 documents are merged into lab3-staging.

* Issues #36 - #42 exist on the GitHub Project board.

SaintCrois approved these changes

Me: Please merge, thank you.

SaintCrois merged commit 42f8406 into lab3-staging

#46 feature/lab3-data-model

Issue 11

Branch: feature/lab3-data-model

Scope

Database

* Add the User model:

- * id
- * name
- * email
- * passwordHash
- * role
- * isActive
- * mustChangePassword
- * timestamps

* Make email unique and case-insensitive.

* Extend Ticket with:

- * ticketOwnerId
- * itPriority
- * problemAppearsResolved
- * resolutionSummary

* Change Ticket.requesterId to reference User.

* Add:

- * PublicComment
- * InternalNote

Migration

Implement the Lab 2 → Lab 3 migration described in specification.md §7.1:

- * Create User records for existing Development Requesters.
- * Backfill Ticket.requesterId.
- * Copy requestedPriority to itPriority.
- * Set problemAppearsResolved = false.
- * Preserve existing Lab 2 ticket data.

Seed

Create idempotent seed data according to the handout:

- * At least 4 active Requesters.
- * At least 1 inactive Requester.
- * At least 3 active IT Staff.
- * At least 1 inactive IT Staff.
- * At least 1 active Administrator.
- * Realistic ticket distribution.
- * Example public comments and internal notes.
- * No sensitive information.
- * Document local-development credentials.

Do not remove client functionality yet.

Tests to Add

- * server/tests/lab-03/unit/email.unit.test.ts
 - * UNIT-07
 - * UNIT-08
- * server/tests/lab-03/migration.api.test.ts
 - * MIG-02
 - * MIG-03
 - * MIG-04
 - * MIG-05

Covers

- * specification.md §7
- * specification.md §7.1

Definition of Done

- * Migration succeeds on the Lab 2 database.
- * Seed is idempotent.
- * All Issue #37 tests pass.
- * Existing Lab 2 data is preserved.

SaintCrois requested changes

SaintCrois:

Looking great! I am not quite sure about uploading the uploads into GitHub. If you would like to

Me:

I have removed the uploaded files and updated .gitignore to exclude the uploads folder.
If there is nothing else left, please proceed to merge.

SaintCrois approved these changes

SaintCrois merged commit 169a961 into lab3-staging

#48 feature/12-lab3-auth

Issue 12

Branch: feature/12-lab3-auth

Scope

Implement the complete authentication, authorization, application shell, and authenticated request

Authentication

- * bcrypt password hashing with cost 12.
- * Password-policy validator.
- * POST /api/auth/login
- * POST /api/auth/logout
- * GET /api/auth/me
- * POST /api/auth/change-password
- * Signed JWT authentication.
- * Store JWT in an httpOnly, Secure, SameSite=Strict cookie.
- * Implement logout/session invalidation.
- * Session middleware populating req.user.
- * Return 401 for unauthenticated requests.
- * Implement X-Requested-With validation for state-changing requests.
- * Remove X-Dev-Requester-Id handling from the server.

Authorization

- * Implement reusable requireRole(...).
- * Implement ticket and attachment ownership checks.
- * Apply authorization rules to existing API routes.
- * Implement the authorization matrix from api-spec.md §6.
- * Ensure correct 401 vs 403 responses.
- * Apply the safe-error response format from api-spec.md §7.

Application Shell

Implement:

- * Login screen.
- * Login validation.
- * Invalid-credential banner.
- * Inactive-account banner.
- * Loading/busy states.

- * Mandatory Change Password screen.
- * Live password-rule checklist.
- * Confirm-password validation.
- * Voluntary Change Password from Profile.
- * Role-based navigation.
- * Profile menu.
- * Logout.
- * Session-expiry handling.
- * Redirect to Login after 401.
- * Shared Forbidden screen.

Requester UI

Remove the Development Requester selector and all related state.

Re-point requester functionality to the authenticated session:

- * My Tickets.
- * Create Ticket.
- * Ticket Detail.
- * Attachments.
- * Public requester actions.

The server must derive the requester identity from req.user.

Confirm that existing Lab 2 requester functionality continues to work for authenticated Requester

Tests to Add

Unit

server/tests/lab-03/unit/password.unit.test.ts

- * UNIT-01
- * UNIT-02

server/tests/lab-03/unit/session.unit.test.ts

- * UNIT-03
- * UNIT-04

API

server/tests/lab-03/auth.api.test.ts

- * API-01-API-07

server/tests/lab-03/authorization.api.test.ts

- * SEC-01-SEC-09

server/tests/lab-03/migration.api.test.ts

- * MIG-01 – full Lab 2 regression under authentication

Authentication UI

client/tests/lab-03/Login.test.tsx

- * UI-01-UI-04

- * UI-26

client/tests/lab-03/ChangePassword.test.tsx

- * UI-05-UI-08

client/tests/lab-03/AppShell.test.tsx

- * UI-24

- * UI-25

Requester UI

client/tests/lab-03/RequesterTicketDetail.test.tsx

- * UI-21-UI-23

- * UI-30

Covers

- * FR-01-04

- * FR-06

- * BR-01-03

- * BR-06-10

- * BR-12

- * BR-13

- * BR-31
- * AC-01-08
- * AC-09-11
- * AC-22
- * AC-23
- * AC-30

Definition of Done

- * Users can log in and log out through the session cookie.
- * Mandatory and voluntary password changes work.
- * Password policies are enforced.
- * Role-based navigation and authorization work correctly.
- * Ticket and attachment ownership is enforced server-side.
- * Correct 401 and 403 responses are returned.
- * Safe-error responses are used for applicable failures.
- * Session expiry redirects users to Login.
- * The shared Forbidden screen is displayed for unauthorized access.
- * Requester screens work using the authenticated session.
- * The Development Requester selector and related state are completely removed.
- * X-Dev-Requester-Id is no longer used anywhere in authentication or requester functionality.
- * Authenticated requester functionality passes the full Lab 2 regression.
- * All Issue #38 tests pass.

SaintCrois approved these changes

Me: Great, thanks. Please merge.

SaintCrois merged commit 7c7c1b4 into lab3-staging

#49 feature/14-lab3-staff-ticket-management

Issue 14

Branch: feature/14-lab3-staff-ticket-management

Scope

Implement the complete IT Staff ticket workflow.

Comments & Internal Notes API

Implement:

- * Public Comment create/list.
- * Internal Note create/list.
- * Content validation.
- * Append-only behavior.
- * Server-controlled author and timestamp.
- * No edit functionality.

Implement:

- * PATCH /api/tickets/:id/resolution

with the required status-gate rule.

Requester Ticket Detail

Add:

- * Public Comments tab.
- * Functional public comment creation/list.
- * Problem Appears Resolved action.
- * Visibility rules.
- * Confirmation step.

IT Staff Ticket Queue

Implement:

- * GET /api/staff/tickets
- * Search.
- * Filters.

- * Sorting.
- * Pagination.
- * Query validation.
- * Empty state.
- * No-results state.
- * Forbidden state.
- * Failure/loading states.
- * Responsive desktop/tablet/mobile layouts.

IT Staff Ticket Detail

Implement:

- * Owner assignment.
- * Priority update.
- * Status update.
- * Eligibility rules.
- * Reassignment rules.
- * Status transition matrix.
- * Resolution Summary.
- * Public Comments.
- * Internal Notes.
- * Attachments.
- * Service Actions placeholder.
- * Confirmation dialog for Resolved/Cancelled.

Tests to Add

Unit

server/tests/lab-03/unit/content.unit.test.ts

- * UNIT-09
- * UNIT-10

server/tests/lab-03/unit/status-transitions.unit.test.ts

- * UNIT-05
- * UNIT-06

server/tests/lab-03/unit/user-ownership.unit.test.ts

- * UNIT-11

API

server/tests/lab-03/comments-notes.api.test.ts

- * API-08

- * API-27-API-34

server/tests/lab-03/staff-queue.api.test.ts

- * API-13-API-17

server/tests/lab-03/staff-ticket-detail.api.test.ts

- * API-18-API-26

UI

client/tests/lab-03/RequesterTicketDetail.test.tsx

- * UI-21-UI-23

- * UI-30

client/tests/lab-03/StaffTicketQueue.test.tsx

- * UI-09-UI-12

- * UI-27

client/tests/lab-03/StaffTicketDetail.test.tsx

- * UI-13-UI-16

- * UI-28

client/tests/lab-03/ZenGreenStyle.test.tsx

- * STYLE-01

- * STYLE-02

Covers

- * FR-09-17

- * BR-04, 05

- * BR-14-25

- * AC-04
- * AC-12-23
- * AC-32

Definition of Done

- * IT Staff can find and manage tickets.
- * Requesters can use public comments.
- * Staff can use internal notes.
- * Ticket ownership, priority, status, and resolution rules are enforced.
- * All Issue #40 tests pass.

SaintCrois approved these changes

Me: Nice, please merge.

Me: Wait, I noticed a UI bug and am fixing it.

Me:

I have fixed the UI issues, including consistency updates and the resolution summary behavior. Please let me know when you are ready for the final review.

SaintCrois approved these changes

SaintCrois merged commit b9f99cf into lab3-staging

#50 feature/lab3-admin-users

Issue 15

Branch: feature/lab3-admin-users

Scope

API

Implement:

- * GET /api/admin/users
- * POST /api/admin/users
- * PATCH /api/admin/users
- * PATCH /api/admin/users/:id/password

Support:

- * Search.
- * Role filtering.
- * User creation.
- * User editing.
- * Duplicate-email detection.
- * Case-insensitive email handling.
- * Self-deactivation protection.
- * Last-Administrator protection.
- * Password reset as initial password.
- * Role and active/inactive state management.

UI

Implement Administrator User Management:

- * User list.
- * Search/filter.
- * Create User slide-over.
- * Edit User slide-over.
- * Field validation.
- * Duplicate-email error.
- * Disabled self-deactivation control.
- * Success notifications.
- * Appropriate loading/error/forbidden states.

Tests to Add

Unit

server/tests/lab-03/unit/user-ownership.unit.test.ts

* UNIT-12

API

server/tests/lab-03/users-admin.api.test.ts

* API-35-API-43

UI

client/tests/lab-03/UserManagement.test.tsx

* UI-17-UI-20

* UI-29

Covers

* FR-18-24

* BR-08

* BR-11

* BR-26-29

* AC-24-30

Definition of Done

- * Administrators can create and manage users.
- * Security restrictions are enforced server-side.
- * User management UI follows the Lab 3 design.
- * All Issue #41 tests pass.

SaintCrois approved these changes

Me: Thank you, please merge.

SaintCrois merged commit 23f00c6 into lab3-staging

#51 feature/16-lab3-integration-qa

Issue 16

Branch: feature/16-lab3-integration-qa

Scope

This issue is the final integration and release milestone.

Migration & Regression

Run the complete migration/regression suite:

- * MIG-01
- * MIG-02
- * MIG-03
- * MIG-04
- * MIG-05

Confirm:

- * Lab 2 data migrated correctly.
- * Seed remains idempotent.
- * No Development Requester selector remains.
- * No X-Dev-Requester-Id references remain.
- * Existing Lab 2 functionality still works under authentication.

E2E Testing

Implement and pass:

- * e2e/lab-03/authentication.spec.ts – E2E-01-E2E-03
- * e2e/lab-03/staff-ticket-flow.spec.ts – E2E-04, E2E-05
- * e2e/lab-03/user-administration.spec.ts – E2E-06-E2E-08
- * e2e/lab-03/responsive.spec.ts – RESP-01-RESP-03

Responsive & Visual QA

Capture screenshots for:

- * Authentication.
- * Staff Ticket Queue.
- * Staff Ticket Detail.

- * User Management.

Store them under:

```
artifacts/lab-03/screenshots/  
├─ authentication/  
├─ staff-queue/  
├─ staff-ticket-detail/  
└─ user-management/
```

Run the Zen Green visual checklist:

- * Design consistency.
- * Role-based navigation.
- * Status/role badges.
- * Editable vs read-only styling.
- * Validation placement.
- * Focus states.
- * Responsive layout.
- * Clipping.
- * Overlap.
- * Overflow.

Complete:

- * STYLE-03

Documentation

Create:

- * docs/lab-03/reviewer.md
- * docs/lab-03/ai-use.md

Include:

- * Reviewer identity.
- * PR links.
- * Review comments.
- * Responses.
- * Approvals.
- * LLM/tools used.
- * 6-10 key prompts.

* My Reflection.

Release

- * Confirm all Definition of Done items in specification.md §10.
- * Ensure all tests in tests.md are Pass.
- * Merge lab3-staging into main.
- * Move Issues #36 – #41 to Done.
- * Verify the final submission evidence directly from main.

Tests to Add / Final Verification

- * All UNIT-* tests.
- * All API-* tests.
- * All UI-* tests.
- * All SEC-* tests.
- * All MIG-* tests.
- * All E2E-* tests.
- * All RESP-* tests.
- * All STYLE-* tests.

Definition of Done

- * main contains the complete Lab 3 implementation.
- * All tests in tests.md show Pass.
- * E2E tests pass against the integrated application.
- * Responsive screenshots are captured.
- * Zen Green visual QA is complete.
- * reviewer.md is complete.
- * ai-use.md is complete.
- * All required submission evidence can be assembled from main.
- * Issue #42 is closed after the final release merge.

SaintCrois:

Great work! Please let me know when you would like it merged.

SaintCrois approved these changes

Me: Thanks.

Pull Requests I reviewed

PR	Branch	Reviewer verdict
#35	feature/11-lab3-specification	Approve
#36	feature/12-lab3-test-plan	Approve
#37	feature/13-user-migration-seed	Approve
#38	feature/14-authentication	Approve
#39	feature/15-role-authorization	Approve
#40	feature/16-requester-regression	Approve
#41	feature/17-staff-ticket-queue	Approve
#42	feature/18-staff-ticket-operations	Approve
#43	feature/19-admin-user-management	Approve
#44	feature/20-lab3-test-implementation	Approve
#45	feature/21-ui-responsive-verification	Approve

[#35](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/35>

[#36](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/36>

[#37](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/37>

[#38](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/38>

[#39](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/39>

[#40](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/40>

[#41](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/41>

[#42](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/42>

[#43](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/43>

[#44](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/44>

[#45](#) <https://github.com/SaintCrois/Mein-Uni-Kampf/pull/45>

#35 feature/11-lab3-specification

Scope

- * Lab 3 specification.
- * UI specification.
- * API contract.
- * Engineering contract for the Lab 3 implementation.

Approval

No review-comment transcript is retained in this repository.

bravefe approved these changes

bravefe merged commit 60f7540 into lab3-staging

#36 feature/12-lab3-test-plan

Scope

- * Lab 3 test plan.
- * Acceptance-criterion traceability.

bravefe requested the test plan review

bravefe:

I saw your test file. Have alot to cover. Please let me know when you want me to proceed.

Me: Thank you! Please merge.

bravefe approved these changes

bravefe merged commit bc619e0 into lab3-staging

#37 feature/13-user-migration-seed

Scope

- * User migration from Lab 2.
- * Deterministic and idempotent seed data.
- * Required active and inactive users.
- * Varied ticket data.

bravefe approved these changes

bravefe merged commit 3f3054f into lab3-staging

#38 feature/14-authentication

Scope

- * Authentication.
- * Mandatory password change.
- * Session handling.
- * Authenticated requester workflow.

bravefe approved these changes

bravefe merged commit ddfc661 into lab3-staging

#39 feature/15-role-authorization

Scope

- * Role authorization.
- * Authenticated application shell.
- * Role-based navigation and access control.

bravefe approved these changes

bravefe merged commit 4cfe035 into lab3-staging

#40 feature/16-requester-regression

Scope

- * Requester regression testing.
- * Public comments.
- * Internal notes.
- * Verification that existing Lab 2 requester functionality works under authentication.

bravefe:

all of the 100, no i mean 101 test are passing so now you may actually work on the ui now

go go go

Me: Alright! Thank you, please merge.

bravefe: gotta go fast

bravefe approved these changes and noted "i forgot to approve but will merge"

bravefe merged commit d4d8435 into lab3-staging

#41 feature/17-staff-ticket-queue

Scope

- * IT Staff ticket queue.
- * Search, filtering, sorting, and pagination.
- * Empty, no-results, forbidden, loading, and failure states.
- * Responsive desktop, tablet, and mobile layouts.

bravefe approved these changes

bravefe merged commit 598d082 into lab3-staging

#42 feature/18-staff-ticket-operations

Scope

- * Staff ticket detail.
- * Ticket ownership assignment.
- * Priority and status updates.
- * Resolution summary.
- * Public comments and internal notes.
- * Attachments and service actions placeholder.

bravefe approved these changes

bravefe merged commit e16ac57 into lab3-staging

#43 feature/19-admin-user-management

Scope

- * Administrator user management.
- * User list, search, and role filtering.
- * User creation and editing.
- * Duplicate-email detection.
- * Self-deactivation and last-Administrator protection.
- * Password reset and active/inactive state management.

bravefe approved these changes

bravefe merged commit 480d046 into lab3-staging

#44 feature/20-lab3-test-implementation

Scope

- * API tests.
- * UI tests.
- * Authorization tests.
- * End-to-end tests.
- * Regression coverage for the integrated Lab 3 workflow.

bravefe approved these changes

bravefe merged commit b5db70e into lab3-staging

#45 feature/21-ui-responsive-verification

Scope

- * Zen Green UI verification.
- * Responsive verification across desktop, tablet, and mobile.
- * Accessibility verification.
- * Screenshot evidence.

bravefe:

The UI verification, responsive screenshots, and documentation are complete and ready for merge.

Me: Thank you! Please merge.

bravefe approved these changes

bravefe merged commit d610341 into lab3-staging

Final Verification Approval Record

The final release verification was prepared on `feature/22-final-lab3-integration` . It records the evidence ready for maintainer review before the final merge to `lab3-staging` and `main` .

Check	Result	Approval / response
Database migration status	3 migrations; schema up to date	Ready for maintainer approval.
Seed data	Idempotent seed completed; required active/inactive users and varied tickets verified	Ready for maintainer approval.
Server API, security, and Lab 2 regression	149 passed, 0 failed	Ready for maintainer

Check	Result	Approval / response
tests		approval.
Client component and regression tests	36 passed, 0 failed	Ready for maintainer approval.
Playwright E2E	12 passed, 0 failed across desktop, tablet, and mobile	Ready for maintainer approval.
Acceptance-criterion traceability	AC-01 through AC-15 mapped in <code>tests.md</code>	Ready for maintainer approval.
UI screenshot evidence	18 screenshots under <code>artifacts/lab-03/screenshots/</code>	Ready for maintainer approval.

Final-PDF Evidence Checklist

- ✅ Part 1 — Engineering specification: `docs/lab-03/specification.md` .
- ✅ Part 2 — API contract: `docs/lab-03/api-spec.md` .
- ✅ Part 3 — UI design and responsive/accessibility record: `docs/lab-03/ui-spec.md` .
- ✅ Part 4 — Test strategy and acceptance-criterion traceability: `docs/lab-03/tests.md` .
- ✅ Part 5 — Authentication, roles, authorization, and requester-regression evidence: `server/client/E2E tests`.
- ✅ Part 6 — IT Staff queue/detail workflow evidence: `server/client/E2E tests and screenshots`.
- ✅ Part 7 — Administrator user-management evidence: `server/client/E2E tests and screenshots`.
- ✅ Part 8 — Desktop, tablet, and mobile screenshot evidence: `artifacts/lab-03/screenshots/` .
- ✅ Part 9 — Review, AI-use, seed, migration, and final release verification: this file and `ai-use.md` .